AI Engineering and Claude Architect · Part II, Retrieval

RAG, Lesson 4

Query Understanding, Context Assembly and Citations

One analyst report says the market is $4.1B, and another says $5.6B. Your assistant picked the larger one, dropped the sources, and stated it as fact. Both numbers were correct, two years apart. Provenance is what would have shown that.

Module	AI05 of 13
Lesson	4 of 7
Prerequisites	AI03 L2, AI03 L6, AI04 L3
Exam relevance	**Direct for provenance.** TS 5.6 in full, and TS 1.3's structured context. This is the one exam-heavy lesson in the module
Reading time	~45 minutes

Contents

1. Query Understanding, Context Assembly and Citations

Lesson 4 of 7 · RAG

Query Understanding, Context Assembly and Citations

Key idea

What happens around retrieval matters as much as the search itself. Rewrite the question so that it can be searched, assemble the results in a deliberate order, and keep every claim tied to its source. **Sources disappear silently during summarization unless your data structures force them to survive, and conflicts must be annotated, not resolved by picking a number.**

> *Before: context layout (AI03 L6), fusion and rerank (AI04 L3). Teaches: query rewriting, decomposition, HyDE, context assembly, the Citations API, claim-source mappings, conflict and date handling, coverage reporting. Used by: AI05 L7, AI06 L6, AI06 L10, AI13 L6.*

The hook: two true numbers, one wrong answer

Your research assistant retrieves two good chunks:

Source	Says	Published
Report A	The market reached $4.1B	2023
Report B	The market is estimated at $5.6B	2025

A summarizing step condenses them, and the final answer reads: "The market is $5.6B." Three things have gone wrong:

1. The source is gone, so nobody can check it.
2. The date is gone, so growth now looks like a contradiction, or vanishes.
3. If the model had picked $4.1B, the answer would be just as confident, and two years stale.

> *Key idea*
>
> **TS 5.6 tests exactly this**: attribution is lost when findings are compressed without claim-source mappings, conflicts need annotating with attribution rather than arbitrary selection, and dates must be required so that temporal differences aren't misread as contradictions.

Intuition: footnotes in a due-diligence report

A good analyst's report doesn't just state figures. Each one has a footnote: who said it, where and when. When sources disagree, the analyst doesn't quietly choose. They write: "Report A says X in 2023, and Report B says Y in 2025. The difference reflects two years of growth." Their reader can then trust it, or check it.

Your pipeline should behave like that analyst. And the footnotes have to travel with the facts through every step.

Part 1: Understanding the query

Users don't write search queries. The question as asked is often a poor match for your documents.

Problem	Example	Technique
Depends on the conversation	"What about for contractors?"	**Rewrite** using history, into a standalone question
Several questions in one	"Compare card and wallet refund times"	**Decompose**, then search each
Different vocabulary	"money back"	Hybrid search handles much of it (AI04 L3)
Very short, or abstract	"failover"	**Expansion**, or HyDE
Implied filters	"last year's policy for India"	**Extract filters** using structured output
Wrong source	"How many tickets?"	**Route** (AI05 L1)

Rewriting

This is the highest-value technique for conversational RAG. Without it, follow-up questions search for almost nothing.

Turn	Text
User, turn 1	"What's our parental leave policy in India?"
User, turn 2	"What about for contractors?"
Rewritten query	"Parental leave policy for contractors in India"

Use a small model at low effort, with a few examples, and **log both forms**, or you won't be able to debug it.

HyDE

HyDE (Gao and colleagues, 2022) has a model write a hypothetical answer, embeds that, and searches for real documents like it. The idea is that an answer looks more like a document than a question does. Its cost is an LLM call per query, and the abstract relies on the encoder filtering out incorrect details. So measure it, and it's rarely a first step.

Should you use any of these?

Each adds latency and cost. **Add one only if your eval, segmented by query type, shows the need.**

Part 2: Assembling the context

Retrieved chunks aren't a prompt yet.

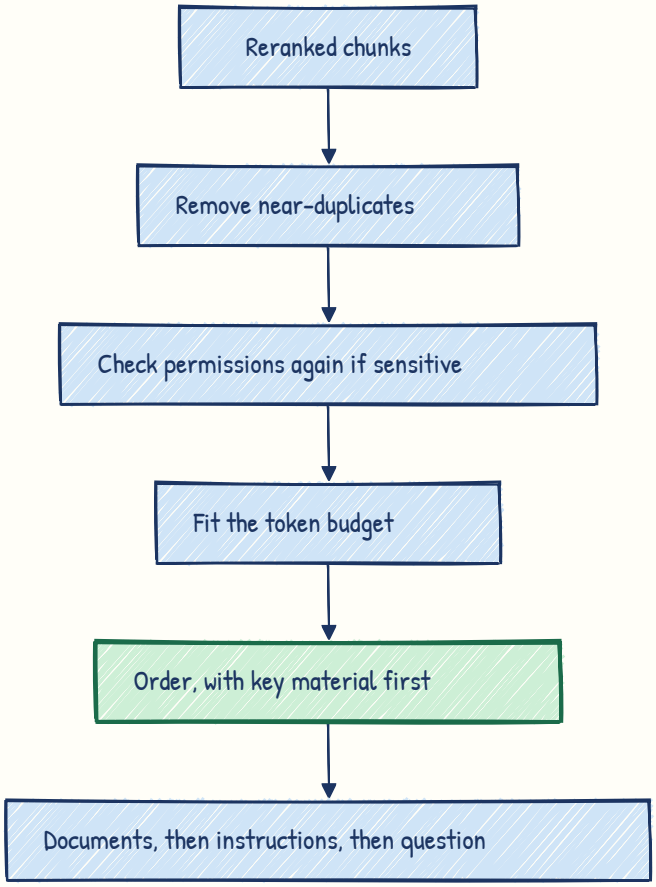


Figure 1. From retrieved chunks to prompt. Deduplication and budgeting come first, then an order chosen deliberately, with long documents above the question.

Step	Why	Source
Deduplicate	Overlap and copies waste budget	AI05 L2
Fit a budget	Fewer, better chunks beat many	HLD M21B
Put documents first, question last	Anthropic reports up to 30% improvement in tests, on complex inputs	AI03 L2
Put best material at the edges of long inputs	Position effects	Lost in the Middle, TS 5.1
Wrap in tags, with source and date	Lets the model cite and compare	AI03 L2
State the rules	Answer only from these. Say so if they're silent	AI02 L6

```
<documents>
  <document index="1">
    <source>refund-policy.md, Refund timing</source>
    <published>2026-03-01</published>
    <document_content>Card refunds take 5 to 10 business days...</document_content>
  </document>
</documents>

Answer using only these documents. If they don't contain the answer, say so.
Where documents disagree, report both, with their sources and dates.

Question: How long do card refunds take?
```

Remember that retrieved text is untrusted, and AI09 L2 covers why.

Part 3: Citations on the Claude API

Why not simply ask for citations in the prompt?

Models can invent plausible citations. Anthropic's Citations feature changes the mechanism. According to its documentation, compared with prompt-based citation:

Benefit	Stated reason
Reliability	The API extracts `cited_text` directly, so citations "are guaranteed to contain valid pointers to the provided documents"
Quality	In Anthropic's evaluations, significantly more likely to cite the most relevant quotes
Cost	`cited_text` doesn't count toward output tokens, or toward input tokens when passed back

Note precisely what's guaranteed: **valid pointers to your documents.** That doesn't guarantee that the cited passage supports the claim. You still need evals for that (AI05 L6).

How it works

Document type	Chunking for citation	Location format
Plain text	Automatically into sentences	Character indices, 0-indexed
PDF	Text extracted, then sentences. Image citations aren't supported	Page numbers, 1-indexed
Custom content	Your blocks, as provided	Block indices, 0-indexed

From `code/ai05-rag/.../CitedAnswers.java`, which compiles against `anthropic-java` 3.2.0, and hasn't been run:

```
for (Chunk chunk : chunks) {
    // Citations must be enabled on all documents in a request, or on none.
    blocks.add(ContentBlockParam.ofDocument(DocumentBlockParam.builder()
            .source(PlainTextSource.builder().data(chunk.text()).build())
            .title(chunk.title())
            .citations(CitationsConfigParam.builder().enabled(true).build())
            .build()));
}
```

> **Citations and structured outputs don't combine**
>
> The documentation states these are **incompatible**, because citation blocks interleave with text, which conflicts with strict JSON schemas. For values with citations, it suggests asking Claude to answer with one line per field, then parsing the lines in your own code. It also warns that a prompt asking for only JSON can result in few or no citations.

Part 4: Provenance through a multi-step pipeline

Citations help at the final answer. In a multi-agent pipeline, attribution has to survive **every step before that.**

Rule 1: Separate content from metadata

Structured findings make it impossible to summarize away the source by accident. From `Provenance.java`:

```
public record Finding(String claimKey, String value, String excerpt, String sourceUrl,
        String published) {
}
```

This is TS 1.3 and TS 5.6 together: use structured data formats to separate content from metadata, including source URLs, document names and page numbers. Downstream agents **must preserve and merge these mappings.**

Rule 2: Annotate conflicts, and don't resolve them silently

```
long distinctValues = evidence.stream()
        .map(Finding::value).distinct().count();
long distinctDates = evidence.stream()
        .map(Finding::published).distinct().count();
String status;
if (distinctValues == 1) {
    status = evidence.size() > 1 ? "WELL_SUPPORTED" : "SINGLE_SOURCE";
} else {
    status = distinctDates > 1 ? "DIFFERS_BY_DATE" : "CONTESTED";
}
```

Real output, on invented data:

```
adoption_growth  CONTESTED        [31% (2025), 18% (2025)]
market_size      DIFFERS_BY_DATE  [$4.1B (2023), $5.6B (2025)]
survey_value     WELL_SUPPORTED   [$5.6B (2025), $5.6B (2025)]
```

Status	What the report does
`WELL_SUPPORTED`	State it, citing both
`DIFFERS_BY_DATE`	Show the trend, with dates. **It's not a contradiction**
`CONTESTED`	Show both, with sources. Don't pick one
`SINGLE_SOURCE`	State it, attributed to that one source

A real system also has to check that sources are actually independent. Two articles quoting one survey aren't two sources.

Rule 3: Verify citations in code

If you're using prompt-based citations, verify them yourself. Real output:

```
real excerpt verified: true
invented excerpt verified: false
```

```
static boolean excerptVerified(Finding finding, Map<String, String> corpus) {
    String source = corpus.get(finding.sourceUrl());
    return source != null && source.contains(finding.excerpt());
}
```

It's a small check that catches invented quotes. Normalize whitespace in real code, since an exact match is brittle. It can't tell whether the excerpt really supports the claim.

Rule 4: Report coverage gaps

From TS 5.3 and the guide's exercises: if a source was unavailable, the report must say so.

Section	Contents
Well-established findings	Multiple independent sources
Contested findings	Conflicts, with attribution
Coverage gaps	Topics with missing or failed sources

Silence looks like an absence of evidence, when it may be an outage.

Rule 5: Render by content type

TS 5.6 adds a final point: render financial data as tables, news as prose, and technical findings as structured lists, rather than forcing everything into one format.

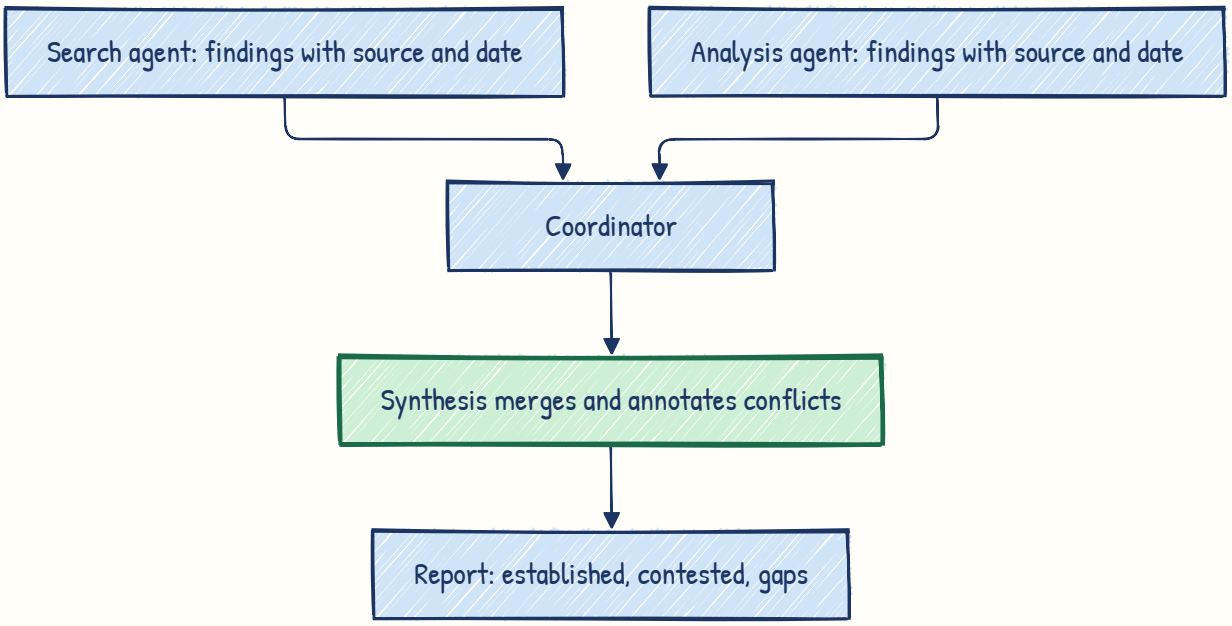


Figure 2. Provenance surviving a pipeline. Each agent passes structured claims forward, and synthesis merges them without dropping sources, dates or conflicts.

How Anthropic solved it

Problem	Anthropic's approach
Hand-rolled citations can be invented	The Citations feature
Long inputs lose focus	Long-document ordering guidance
Synthesis loses sources	Anthropic's multi-agent research system is covered in AI06 L6, once that post has been read
Exam-level expectations	TS 5.6

The production architecture

This is a recommended reference design, not a specific company's system.

Baseline: raw question, raw chunks. Follow-up questions fail, and sources vanish.

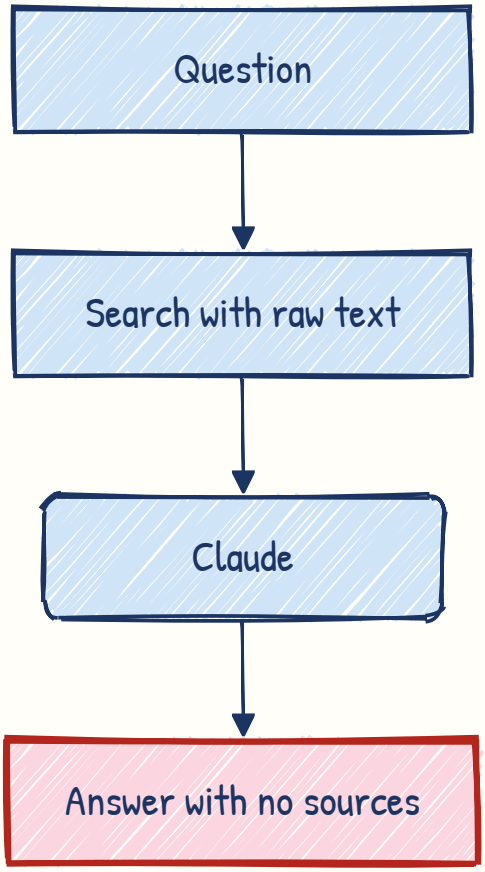


Figure 3. Baseline. The raw question searches badly, and the answer can't be checked.

Improved: rewriting and API citations. Answers can now be checked. New problem: **citations prove that a pointer is valid, not that the claim is supported**, and conflicts can still be flattened.

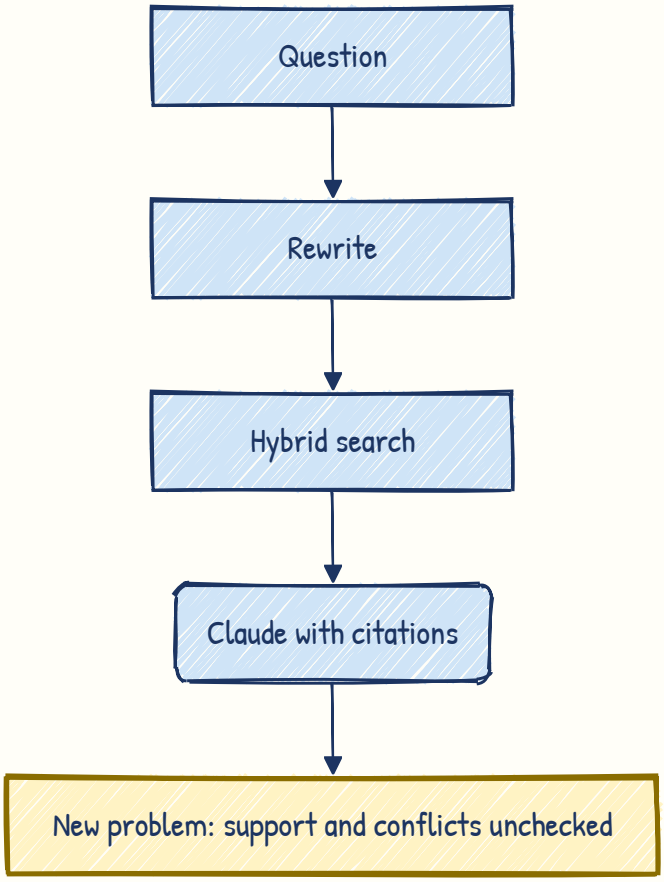


Figure 4. Improved. Better queries and checkable sources, but no check that sources support the claims, or that conflicts survive.

Production: structured evidence, with checks and sampled review.

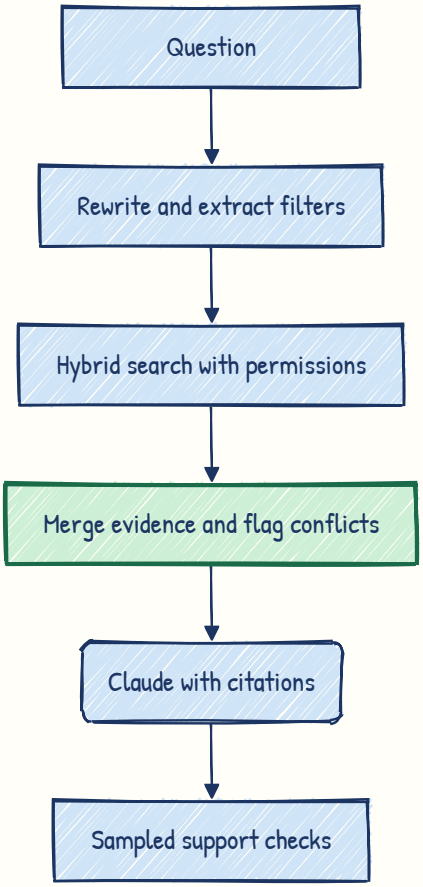


Figure 5. Production. Evidence stays structured throughout. Code merges conflicts before generation, the model cites, and sampled evaluation checks that citations really support the claims.

Failures

Failure	Containment
Rewrite changes the question's meaning	Log both, and show the interpreted query to the user
Citation points to a retrieved chunk that doesn't support the claim	Sampled evaluation (AI05 L6)
All sources failed	Say so, and report the gap
Old document outranks a new one	Date metadata, filtered or boosted
Two sources repeat one origin	Dedupe by origin, where it's known

Trade-offs

Decision	Gain	Cost
Query rewriting	Follow-ups work	A model call, and possible distortion
API citations	Valid pointers, efficient output	Can't combine with structured outputs
Structured findings	Provenance survives	More schema and tokens
Showing conflicts	Honest, and auditable	Longer answers, and users must read them

When you can skip most of this: a single-document assistant that isn't conversational.

Common mistakes

1. **Searching with the raw follow-up question.** "What about contractors?" finds little.
2. **Summarizing without mappings.** Sources disappear.
3. **Picking one number when sources disagree.** Annotate instead.
4. **Treating date differences as contradictions.** Require dates.
5. **Assuming a valid citation proves the claim.** It proves the pointer.
6. **Combining citations and `output_config.format`.** They're incompatible.
7. **Hiding failed sources.** Report coverage gaps.

What to say in an interview

"Provenance has to be structural. Each finding carries its excerpt, source and date, and synthesis merges those rather than summarizing them away. Conflicts get annotated. Different dates become a trend, and same-date disagreements are reported as contested. Claude's citations guarantee valid pointers, though not support, so I'd sample claims in evaluation."

Certification lens

How this lesson maps to CCAR-F

Direct: all of **TS 5.6**, plus parts of **TS 1.3**, **TS 5.1** and **TS 5.3**. Scenario 3 (research system) uses these heavily. Wrong answers tend to pick one value, average them, drop the source, or hide a failure.

These are original practice questions, not real or leaked exam questions.

Q Q1 · Exam style, scenario: Multi-Agent Research System · Medium. Final reports read well, but claims can no longer be traced to sources. Subagents return prose summaries. What's the most effective fix?

A. Ask the synthesis agent to add citations from its own knowledge.

B. Run a final agent that searches the web for a source for each claim.

C. Require subagents to output structured claim-source mappings that downstream agents must preserve.

D. Increase the synthesis agent's context window.

A **C** (TS 5.6). A invites invented citations. B may attach a source that didn't support the original finding. D doesn't change what the summaries contain.

Q Q2 · Exam style, scenario: Multi-Agent Research System · Medium. Two credible sources give different figures for the same metric, in the same year. What should the document analysis subagent do?

A. Keep the more recent publication's figure.

B. Average the two figures.

C. Keep the figure from the larger organization.

D. Include both values with attribution, and let the coordinator decide how to reconcile them.

A **D** (TS 5.6). A, B and C all silently discard or invent information, and the guide says to annotate conflicts rather than select arbitrarily.

Q Q3 · Exam style, scenario: Multi-Agent Research System · Hard. Choose the **two** changes that best prevent temporal differences from being misread as contradictions, and preserve context.

A. Require publication or collection dates in subagent outputs.

B. Include methodological context alongside findings.

C. Convert all numbers to the most recent year's estimates.

D. Drop findings older than two years.

E. Keep only the highest number for each metric.

A **A and B** (TS 5.1, TS 5.6). C invents data. D loses history. E misrepresents evidence.

Q Q4 · Exam style, scenario: Multi-Agent Research System · Medium. One search subagent timed out, so a section has less evidence. How should the final report handle this?

A. Omit the topic without comment.

B. Fill the section using the model's general knowledge.

C. Mark the section with a coverage annotation stating which sources were unavailable.

D. Fail the whole report.

A **C** (TS 5.3). A hides the gap. B risks fabrication. D discards the useful work already done.

Q Q5 · Configuration · Medium. A request enables citations on one document and not on another, and also sets `output_config.format`. What are the two problems?

A (1) Citations must be enabled on all documents or on none. (2) Citations are incompatible with structured outputs. For cited structured values, ask for one line per field, and parse it in your own code.

Revision - Lesson 4

→ Rewrite follow-up questions into standalone queries, and log both forms.

→ Decompose, use HyDE or extract filters only if evals by query type show the need.

→ Assemble deliberately: dedupe, budget, then documents first and the question last.

→ API citations guarantee valid pointers. They don't prove support.

→ cited_text doesn't count toward output tokens.

→ Citations must be on all documents or none, and they don't combine with structured outputs.

→ Provenance needs structure: claim, excerpt, source and date travel together.

→ Annotate conflicts with attribution. Don't pick, average or drop.

→ Require dates, so that growth isn't misread as contradiction.

→ Report coverage gaps. Silence may be an outage.

→ Render by content type: tables, prose, lists.

Your notes